

Panel Estimating Effects of Macroeconomic Determinants on GDP Growth in ASEAN Countries

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Abstract

Economic growth remains a crucial policy goal for ASEAN countries, each with unique macroeconomic structures and developmental pathways. Understanding the domestic macroeconomic factors that influence GDP growth is key to formulating effective and sustainable policies. While the impact of external variables like foreign direct investment and trade openness has been widely explored, there is limited focus on how internal factors—such as inflation, government spending, interest rates, and labor force participation—affect GDP growth within the ASEAN context. This study fills this gap by examining the influence of domestic macroeconomic determinants on GDP growth in ASEAN countries, using a panel data approach covering the period from 1991 to 2023. The study utilizes World Bank data for ten ASEAN nations and includes variables such as GDP per capita, inflation, unemployment, and exchange rates. The results indicate that unemployment, exchange rates, and GDP per capita significantly influence economic growth. The Arellano-Bover/Blundell-Bond model reveals that lagged GDP growth, along with GDP per capita and inflation, have positive effects on growth, while unemployment shows a positive but unexpected relationship. These findings provide valuable insights for policymakers seeking to enhance economic stability and growth in the ASEAN region by focusing on domestic economic indicators.

Keywords : Economic growth, ASEAN countries, macroeconomic determinants, GDP growth, panel data analysis, inflation, unemployment, exchange rate, GDP per capita, domestic macroeconomic factors, policy analysis.

1. Introduction

Economic growth remains a key policy objective for countries in the ASEAN region, which are characterized by diverse macroeconomic structures and development paths. Identifying the domestic macroeconomic factors that influence GDP growth is essential for designing effective and sustainable economic policies. While global economic literature offers numerous insights into the impact of international variables such as foreign direct investment and trade openness, relatively limited attention has been given to how internal macroeconomic indicators such as inflation, government spending, interest rates, and labor force participation affect economic performance specifically within the ASEAN context.

To date, no comprehensive panel-based study has been conducted that exclusively investigates the influence of domestic macroeconomic variables (excluding external factors) on GDP growth across ASEAN countries. This lack of focused empirical research represents a significant gap, especially considering the increasing need for region-specific policy

analysis in light of post-pandemic recovery efforts and macroeconomic uncertainty. Existing studies on macroeconomic determinants of growth tend to focus more on external factors or aggregate models that combine domestic and international variables, often leaving a specific gap in understanding how each internal factor uniquely influences GDP growth within the ASEAN region (Le, 2021; Tran & Doan, 2022).

This study aims to estimate the effects of selected domestic macroeconomic determinants on GDP growth in ASEAN countries using panel data analysis. By focusing solely on internal indicators and excluding international economic flows, this research offers a more precise view of how domestic policy levers relate to economic growth. The findings are expected to provide valuable evidence for national policymakers aiming to strengthen macroeconomic stability and growth prospects in the region.

2. Literature Review

In recent decades, panel data analysis has become a dominant approach in evaluating macroeconomic determinants of economic growth across countries. This study utilizes the panel data framework to capture the complex economic dynamics in the ASEAN region, which consists of countries with varying levels of development and economic structures. The use of panel data allows for controlling individual heterogeneity and detecting temporal dynamics that are difficult to identify through time-series or cross-sectional approaches alone (Baltagi, 2005; Hsiao, 2014). Amid the challenges and opportunities of economic integration, this approach is highly relevant in understanding the contributions of inflation, trade, investment, and other macroeconomic variables to Gross Domestic Product (GDP). The strength of this approach also lies in its ability to explore both long-term and short-term relationships between variables, providing a stronger empirical foundation for regional economic policy-making.

Furthermore, literature suggests that the ASEAN region has become a natural laboratory for testing the hypothesis of convergence growth and economic integration. The ADB (2020) highlights that capital flows, technology transfers, and intra-regional trade drive more inclusive growth in ASEAN's developing countries. In this context, the panel data approach plays a crucial role in uncovering the differential effects of policies across countries and time, providing a sharper understanding of how economic determinants work within different institutional and structural contexts. Barro and Sala-i-Martin (1995) also emphasize that economic growth models combined with panel data can explain variations in economic performance across countries by taking into account institutional factors, infrastructure, and human capital accumulation. Therefore, panel analysis not only provides empirical results but also opens up space for deep theoretical reflection on the dynamics of development in ASEAN.

3. Data and Methodology

3.1 Data Set

This study uses annual panel data from 1991 to 2023 to analyze the impact of macroeconomic determinants on Gross Domestic Product (GDP) growth in ASEAN member countries. The selection of the period 1991–2023 is based on the availability of consistent and complete data for all the variables used in the study, as well as covering a sufficiently long period to observe the structural economic dynamics in the ASEAN region. The data used covers ten ASEAN countries: Indonesia, Malaysia, Myanmar, Thailand, the Philippines, Vietnam, Singapore, Brunei Darussalam, Laos, and Cambodia.

The primary data source for this study is the World Bank Open Data, which provides official and verified macroeconomic data for countries around the world. The reliability and completeness of the World Bank's database make it an ideal source for cross-country panel data analysis.

This study involves one dependent variable and four independent (explanatory) variables, all of which are macroeconomic in nature and have been frequently used in development economics literature. These variables are explained as follows:

- **GDP Growth (annual %):** Used as the main dependent variable in this study, representing the rate of economic growth of a country from year to year. This variable was chosen because it is the primary indicator for assessing a country's macroeconomic performance.
- **GDP per Capita (constant 2015 US\$):** Represents the average economic welfare level of the population. This variable is used to observe how prosperity levels affect economic growth from the demand side and productivity.
- **Inflation Rate (Consumer Prices, annual %):** Reflects the rate of increase in the general price level of goods and services. High inflation can disrupt economic stability, while moderate inflation can signal domestic demand growth.
- **Unemployment Rate (% of total labor force):** Depicts the level of unemployment in a country and serves as an important indicator for assessing the utilization of labor as one of the production factors.
- **Exchange Rate (official, LCU per US\$, period average):** This variable indicates the exchange rate of the local currency against the US dollar. Fluctuating exchange rates often impact trade stability, investment, and the competitiveness of exports and imports in a country.

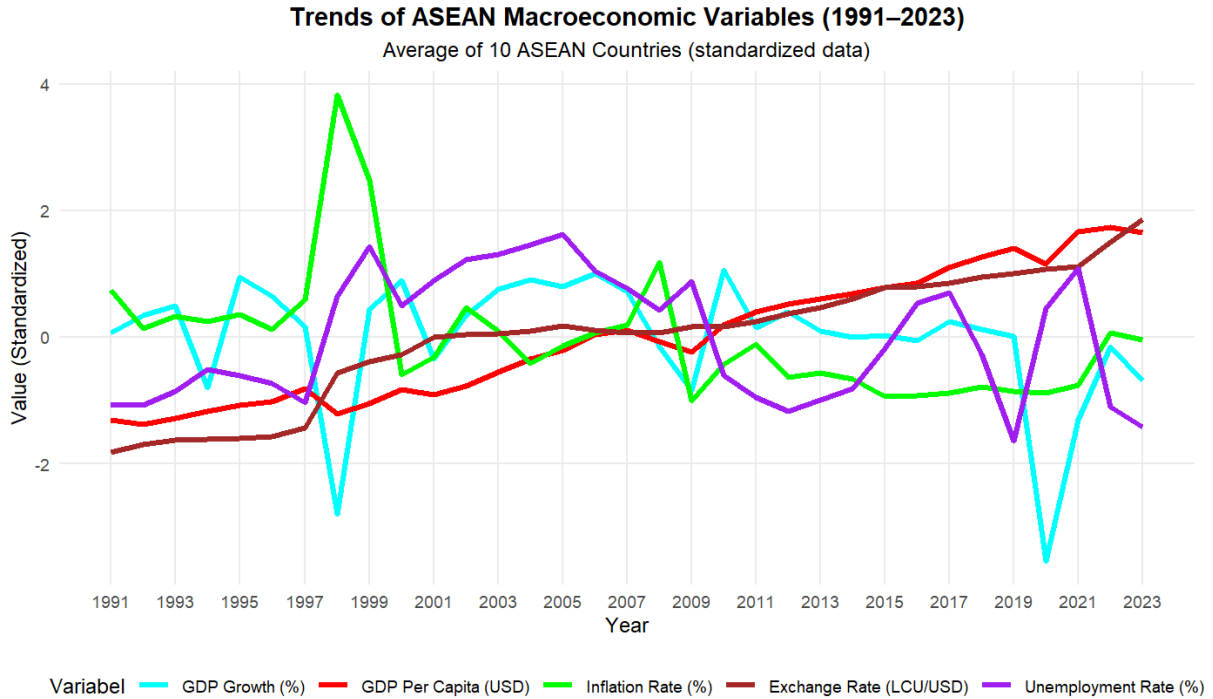


Figure 1. Trends of ASEAN Macroeconomic Variables (1991–2023)
Average of 10 ASEAN Countries (standardized data)

The following graph displays the standardized average trends of five key macroeconomic variables across ten ASEAN countries over the period from 1991 to 2023. The graph illustrates the annual fluctuation dynamics of GDP growth, GDP per capita, inflation, unemployment rate, and exchange rate against the US dollar. The data in the graph have been standardized (in Z-score form) to facilitate comparison among variables with different units and scales.

The graph shows that some variables experienced sharp fluctuations, particularly GDP growth (light blue) and inflation (green), reflecting macroeconomic volatility due to financial crises such as the Asian Financial Crisis in 1997–1998 and the COVID-19 pandemic in 2020. In contrast, variables such as GDP per capita (dark red) and exchange rate (brown) display more stable and generally upward trends over the long term. Meanwhile, the unemployment rate (purple) also shows significant fluctuations at certain points in time, which correlate with global and regional economic shocks.

3.2 Methodology

In this study, the data used includes various macroeconomic indicators from ASEAN countries over a specific period of time. The initial process involved identifying and selecting variables considered relevant in influencing economic growth (GDP Growth), which is then defined as the dependent variable. The explanatory variables used include GDP per Capita (GDP_Percapita), inflation rate (Inflation), unemployment rate (Unemployment), and

exchange rate against the US dollar (ExchangeRate). Table 1 below presents the naming and acronyms of each variable used in the analysis.

Table 1. Variable descriptions and expected signs			
Variabel		Penamaan	Akronim
Dependent variable	GDP Growth	GDP Growth (%)	GDP_Growth
Explanatory variables	Selected-specific factor	GDP per Capita (\$) Inflation rate (%) Unemployment rate (%) Exchange rate (LCU/USD)	GDP_Percapita Inflation Unemployment Kurs

Once the data were collected and organized, the next step involved conducting a preliminary exploration of each variable using descriptive statistics. This step was essential to understand the distribution and general characteristics of the dataset. Table 2 presents the output of the “*summary()*” function in R, which provides key information such as the number of observations (Obs), minimum value (Smallest), first quartile (Q1), median (Q2), third quartile (Q3), maximum value (Largest), and mean.

Table 2 shows that there are 330 observations for each variable. Economic growth (GDP_Growth), for example, exhibits substantial variation, ranging from –34.809 to 14.520, with an average of 4.989. Likewise, GDP per capita varies significantly, from as low as 166.7 to as high as 67,948.9 USD, with a mean of approximately 10,249.6 USD. Other variables such as inflation, unemployment, and exchange rate also display considerable variation, reflecting the economic heterogeneity across countries and over time within the scope of this dataset.

Table 2. Summary of descriptive statistics							
	Obs	Smallest	Q1	Q2	Q3	Largest	Mean
GDP_Growth	330	-34.809	3.706	5.735	7.384	14.520	4.989
GDP_Percapita	330	166.7	1365.2	2651.7	7642.8	67948.9	10249.6
Inflation	330	-2.315	1.636	3.740	6.627	125.272	6.418

Unemployment	330	0.119	1.123	2.706	3.854	9.316	2.879
Kurs	330	1.25	3.79	44.26	7650.43	23787.32	3803.82

The results of the correlation analysis are presented in Table 3, which displays the Pearson correlation coefficients among the variables. Overall, the correlations do not indicate any extremely high relationships (approaching ± 1), suggesting no strong indication of multicollinearity. For instance, GDP growth exhibits a moderate negative correlation with unemployment (-0.31), and weak correlations with other variables such as GDP per capita (-0.22), inflation (0.06), and the exchange rate (0.11).

On the other hand, GDP per capita shows a moderate positive correlation with unemployment (0.52), yet this value remains within an acceptable range for further analysis. These findings justify the inclusion of all variables in the model, as there is no evidence of severe multicollinearity. Consequently, all variables are retained for subsequent model testing and estimation.

Table 3. Correlation matrix					
	GDP_Growth	GDP_Percapita	Inflation	Unemployment	Kurs
GDP_Growth	1.00	-0.22	0.06	-0.31	0.11
GDP_Percapita	-0.22	1.00	-0.24	0.52	-0.33
Inflation	0.06	-0.24	1.00	-0.17	0.07
Unemployment	-0.31	0.52	-0.17	1.00	-0.13
Kurs	0.11	-0.33	0.07	-0.13	1.00

4. Result and Discussion

4.1 Unit Root Test

Stationarity testing is crucial in both time series and panel data analysis, as most estimation techniques including panel regression models assume that the variables involved are stationary. Therefore, to ensure the validity of the model, unit root tests were conducted on all variables to be analyzed.

A unit root test is a statistical method used to determine whether a time series variable is stationary or non-stationary. A stationary time series has a constant mean, variance, and autocovariance over time, which is essential for reliable econometric modeling. In contrast, a non-stationary series can produce misleading regression results due to spurious correlations. The presence of a unit root indicates non-stationarity, while its absence suggests that the series is stationary. Common unit root tests include the Augmented Dickey-Fuller (ADF) test, the Phillips-Perron (PP) test, and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test.

Stationarity is crucial in macroeconomic time series analysis because many econometric techniques require the variables to be stationary to ensure valid inference. If a variable is found to have a unit root, it is typically differenced to achieve stationarity.

Based on Table 4, the unit root test results for all variables in the study indicate that they are non-stationary at level form but become stationary after taking the first difference. This conclusion is supported by both the Augmented Dickey-Fuller (ADF) test and the Phillips-Perron (PP) test statistics, which are significant at the 1% level after differencing. Hence, all variables are integrated of order one, $I(1)$, and are suitable for further analysis such as cointegration testing and vector error correction modeling (VECM).

Tabel 4. Unit root test						
Variabel	Levin-Lin-Chu (LLC)		Pasaran-Shin (IPS)		Fisher ADF	
	At level	1st difference	At level	1st difference	At level	1st difference
	Statistic and P-value		Statistic and P-value		Statistic and P-value	
GDP_Growth	-5.7978 3.359e-09	-15.605 2.2e-16	-7.0819 7.109e-13	-17.98 2.2e-16	110.81 1.397e-14	408.24 2.2e-16
GDP_Percapita	3.4955 0.9998	-6.9339 2.048e-12	6.6977 1	-8.7188 2.2e-16	1.7596 1	155.28 2.2e-16
Inflation	-5.1408 1.368e-0	-15.284 2.2e-16	-5.9305 1.51e-09	-18.157 2.2e-16	83.336 1.058e-09	415.86 2.2e-16

	7					
Unemployment	-1.2293 0.1095	-8.4155 2.2e-16	-1.3632 0.08641	-10.62 2.2e-16	28.202 0.1047	196.36 2.2e-16
Kurs	-0.56118 0.2873	-8.4362 2.2e-16	-0.10941 0.4564	-8.6044 2.2e-16	36.419 0.01373	159.05 2.2e-16

4.2 Empirical results and discussion

When comparing the Fixed Effects model with the Arellano-Bover/Blundell-Bond (AB/BB) approach, the assessment is not solely based on coefficient values or statistical significance, but also on how well each model fits the data characteristics and the underlying assumptions of the respective methods. The Fixed Effects model is typically applied to static panel data and is effective in controlling for unobserved individual heterogeneity; however, it lacks the capacity to address temporal dynamics and potential endogeneity among variables. In contrast, the AB/BB model is specifically designed for dynamic panel data, incorporating lagged dependent variables as predictors, thus enabling it to address endogeneity issues through the use of internal instrumental variables.

In this context, choosing the appropriate model also requires considering diagnostic test results, such as the second-order autocorrelation test (AR(2)) to ensure the absence of autocorrelation in the residuals of the dynamic model, and the Sargan test to verify the validity of the instruments. If these tests indicate that the dynamic model satisfies its assumptions, the AB/BB estimator is deemed more appropriate—particularly in cases where the dependent variable, such as GDP growth, is influenced by its past values. Therefore, the comparison between these two models is carried out not only from a statistical perspective but also through methodological and econometric lenses, in alignment with the structure and analytical needs of the data.

Table 5. Estimation Results of the Fixed Effect Model and the Arellano-Bover/Blundell-Bond Estimation

Variabel	Fixed Effect Model	Arellano-Bover/Blundell Bond Estimation
GDP_Growth		GDP Growth L1. 0.32231725 0.003104 **
GDP_Percapita	0.00188818 7.160e-11 ***	0.00143712 0.005165 **
Inflation	0.02655537	0.05326224

	0.258335	0.009091 **
Unemployment	-2.35907855 2.044e-08 ***	1.38890050 0.022164 *
Kurs	-0.00147755 0.000116 ***	-0.00090356 0.010704 *
Observation	320	320
R-sq/Adjusted R-sq Within Between Overall	0.2747252 / 0.2439129 0.6279360 / 0.3302848 0.2569820 / 0.2475469	
F-Test	F-statistic: 28.9773 P < 2.22e-16	
AR(2) Test		Z = 0.3696214, P = 0.71166)
Durbin-Watson Test	DW = 1.5629	
Breusch-Pagan Test	BP = 1.7194	
Kao test for co-integration (Approximation)		Dickey-Fuller = -8.7892 P < 0.01
Sargan test J-test		chisq(440) = 10, P = 1

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

This analysis employs both the Fixed Effects Model and the Arellano–Bover/Blundell–Bond Estimation to examine the influence of macroeconomic variables on GDP growth. The Fixed Effects model indicates that unemployment has a negative and statistically significant effect on GDP growth (-2.3591 , $p < 0.001$), as does the exchange rate (-0.00148 , $p < 0.001$). Meanwhile, GDP per capita shows a positive but relatively small effect (0.00189 , $p \approx 0.0014$), and inflation is found to be statistically insignificant.

The Arellano–Bover/Blundell–Bond model reveals that lagged GDP growth is significantly and positively associated with current growth (0.3223 , $p < 0.01$), reflecting persistence in economic performance. Both GDP per capita (0.00517 , $p < 0.01$) and inflation (0.0533 , $p < 0.01$) have positive and statistically significant effects. Interestingly, unemployment also shows a significant but positive relationship (1.3889 , $p < 0.05$), suggesting possible structural

effects. The exchange rate remains negatively and significantly associated with GDP growth ($-0.00090, p < 0.05$).

The validity of the GMM model is supported by the AR(2) test ($p = 0.711$), the Sargan test ($p = 1.000$), and the Kao test, which indicates the presence of a long-run relationship (ADF = $-8.7892, p < 0.01$). These findings underscore the importance of effective labor market management, exchange rate stability, and income dynamics in driving economic growth.

5. Conclusion

This study demonstrates the differential effects of monetary and fiscal policy tools on key macroeconomic indicators. The simulation results suggest that monetary expansion is effective in boosting short-term economic activity and lowering unemployment. However, this comes at the cost of increased inflation. On the other hand, fiscal contraction appears to suppress inflation but adversely affects output and employment levels.

Given these dynamics, the results underscore the necessity for a balanced and coordinated macroeconomic policy framework. Policymakers must weigh short-term objectives against long-term stability, considering not only the desired economic outcomes but also the unintended side effects of each policy choice. Future studies may benefit from extending this simulation approach to include open economy elements or external shocks, thereby enhancing its policy relevance in real-world settings.

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